

1. Arithmetic Mean Basics: Calculation

The **Arithmetic Mean** (or simply the **Average**) is the sum of all values in a dataset divided by the total number of values.

$$\text{Mean}(\bar{x}) = \frac{\sum x_i}{N}$$

1A. For a Simple List of Numbers (Individual Data)

Example 1: Finding the Mean of a Set

Problem: Find the arithmetic mean of the following five numbers: 12, 18, 25, 10, 35.

Step	Calculation	Result
1. Sum of Values ($\sum x_i$)	$12 + 18 + 25 + 10 + 35$	100
2. Total Count (N)	Number of values	5
3. Calculate Mean ($\bar{x}$)	$\frac{100}{5}$	20

The arithmetic mean is **20**.

or

Example 2: Finding the Mean of Prime Numbers

Problem: Find the arithmetic mean of the first six prime numbers.

Step	Calculation	Result
1. Identify Values (x_i)	The first six prime numbers are 2, 3, 5, 7, 11, 13.	

Step	Calculation	Result
2. Sum of Values ($\sum x_i$)	$2 + 3 + 5 + 7 + 11 + 13$	41
3. Total Count (N)	Number of values	6
4. Calculate Mean ($\bar{x}$)	$\frac{41}{6}$	6.83 (approx.)

The arithmetic mean is $\frac{41}{6}$ or approximately **6.83**.

or

1B. For a List with Frequencies (Discrete Data)

The formula for mean with frequency is:

$$\text{Mean}(\bar{x}) = \frac{\sum f_i x_i}{\sum f_i}$$

Example 3: Direct Method

Problem: The number of books (x_i) read by students in a week, along with the frequency (f_i) of students who read that number, is given. Find the mean number of books read.

Number of Books (x_i)	Number of Students (f_i)	$f_i x_i$
2	3	$2 \times 3 = 6$
4	2	$4 \times 2 = 8$
5	1	$5 \times 1 = 5$
1	2	$1 \times 2 = 2$
Total	$\sum f_i = 8$	$\sum f_i x_i = 21$

$$\text{Mean}(\bar{x}) = \frac{\sum f_i x_i}{\sum f_i} = \frac{21}{8} = 2.625$$

The mean number of books read is **2.625**.

Example 4: Shortcut (Assumed Mean) Method

Problem: Find the mean number of siblings using the Assumed Mean Method.

Siblings (x_i)	Households (f_i)
0	2
1	3
2	4
3	1

- **Assumed Mean (A):** Let's choose $A = 1$.
- **Formula:** $\bar{x} = A + \frac{\sum f_i d_i}{\sum f_i}$, where $d_i = x_i - A$.

x_i	f_i	Deviation $d_i = x_i - 1$	$f_i d_i$
0	2	$0 - 1 = -1$	$2 \times (-1) = -2$
1	3	$1 - 1 = 0$	$3 \times 0 = 0$
2	4	$2 - 1 = 1$	$4 \times 1 = 4$
3	1	$3 - 1 = 2$	$1 \times 2 = 2$
Total	$\sum f_i = 10$		$\sum f_i d_i = 4$

$$\text{Mean}(\bar{x}) = 1 + \frac{4}{10} = 1 + 0.4 = 1.4$$

The mean number of siblings is **1.4**.

1C. For Data in Ranges (Continuous Data / Grouped Frequency Distribution)

The mean is calculated using the class mark (x_i , or midpoint) of each class interval.

Example 5: Direct Method (for Grouped Data)

Problem: Find the mean marks from the following distribution.

Marks (Class Interval)	Frequency (f_i)
0-10	5
10-20	12
20-30	15
30-40	6
40-50	2

- **Formula:** $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$, where x_i is the class mark (midpoint).

Marks	f_i	Class Mark x_i	$f_i x_i$
0-10	5	$\frac{0+10}{2} = 5$	$5 \times 5 = 25$
10-20	12	$\frac{10+20}{2} = 15$	$12 \times 15 = 180$
20-30	15	$\frac{20+30}{2} = 25$	$15 \times 25 = 375$
30-40	6	$\frac{30+40}{2} = 35$	$6 \times 35 = 210$
40-50	2	$\frac{40+50}{2} = 45$	$2 \times 45 = 90$
Total	$\sum f_i = 40$		$\sum f_i x_i = 880$

$$\text{Mean}(\bar{x}) = \frac{880}{40} = 22$$

The mean marks is **22**.



Example 6: Step Deviation Method

Problem: Calculate the mean daily wage (in \$) using the Step Deviation Method.

Daily Wage (\$)	Frequency (f_i)
100-120	3
120-140	5
140-160	8
160-180	4

- **Class Size (h):** $120 - 100 = 20$.
- **Assumed Mean (A):** Let's choose the class mark of the third interval: $A = 150$.
- **Formula:** $\bar{x} = A + \left(\frac{\sum f_i u_i}{\sum f_i} \right) \times h$, where $u_i = \frac{x_i - A}{h}$.

Wage	f_i	Class Mark x_i	$d_i = x_i - 150$	$u_i = d_i/20$	$f_i u_i$
100-120	3	110	-40	-2	-6
120-140	5	130	-20	-1	-5
140-160	8	150	0	0	0
160-180	4	170	20	1	4
Total	$\sum f_i = 20$				$\sum f_i u_i = -7$

$$\text{Mean}(\bar{x}) = 150 + \left(\frac{-7}{20} \right) \times 20$$

$$\text{Mean}(\bar{x}) = 150 + (-7) = 143$$

The mean daily wage is \$\$\$143.



2. Special Cases & Other Types of Means

2A. Finding a Missing Number

Example 7: Single Missing Value

Problem: The arithmetic mean of 5 numbers is 45. Four of the numbers are 30, 65, 40, and 52. What is the fifth number?

1. **Find the Total Sum:** The sum of N numbers is $N \times \text{Mean}$.

$$\text{Total Sum} = 5 \times 45 = 225$$

2. **Find the Sum of Known Numbers:**

$$\text{Sum of Known} = 30 + 65 + 40 + 52 = 187$$

3. **Find the Missing Number:**

$$\text{Missing Number} = \text{Total Sum} - \text{Sum of Known}$$

$$\text{Missing Number} = 225 - 187 = 38$$

The fifth number is **38**.



Example 8: Score in the Last Game

Problem: A cricket player's average score in 8 innings is 58 runs. Their scores in the first 7 innings are 45, 72, 60, 50, 81, 35, and 68. What was the score in the 8th inning?

1. **Total Runs in 8 Innings:**

$$\text{Total Runs} = 8 \times 58 = 464$$

2. **Total Runs in 7 Innings:**

$$\text{Sum of 7 Scores} = 45 + 72 + 60 + 50 + 81 + 35 + 68 = 411$$

3. **Score in the 8th Inning:**

$$\text{Score} = 464 - 411 = 53$$

The score in the 8th inning was **53 runs**.



2B. Combined Average

The **Combined Average** is the overall average of two or more groups. It is a type of weighted average where the weights are the number of items in each group.

$$\text{Combined Average} = \frac{N_1\bar{x}_1 + N_2\bar{x}_2}{N_1 + N_2}$$

Example 9: Combined Average of Two Classes

Problem: Section A has 30 students (N_1) with an average score of 75 ($\bar{x}_1$). Section B has 20 students (N_2) with an average score of 85 ($\bar{x}_2$). Find the combined average score of the entire class.

1. **Total Score of Section A:** $N_1\bar{x}_1 = 30 \times 75 = 2250$
2. **Total Score of Section B:** $N_2\bar{x}_2 = 20 \times 85 = 1700$
3. **Total Students:** $N_1 + N_2 = 30 + 20 = 50$
4. **Combined Average:**

$$\text{Combined Average} = \frac{2250 + 1700}{50} = \frac{3950}{50} = 79$$

The combined average score is **79**.



Example 10: Combined Average Weight

Problem: The average weight of 10 boys (N_B) is 40 kg ($\bar{x}_B$), and the average weight of 5 girls (N_G) is 35 kg ($\bar{x}_G$). What is the average weight of all 15 students?

1. **Total Weight of Boys:** $10 \times 40 = 400$ kg
2. **Total Weight of Girls:** $5 \times 35 = 175$ kg
3. **Total Students:** $10 + 5 = 15$
4. **Combined Average:**

$$\text{Combined Average} = \frac{400 + 175}{15} = \frac{575}{15} \approx 38.33$$

The average weight of all 15 students is approximately 38.33 kg.



2C. Weighted Average

The **Weighted Average** is used when different data points have different degrees of importance (weights, w_i).

$$\text{Weighted Average} = \frac{\sum w_i x_i}{\sum w_i}$$

Example 11: Weighted Grade Point Average (WGPA)

Problem: A student has the following grades (x_i): Math (4 credits, $w_i = 4$, grade 90), English (3 credits, $w_i = 3$, grade 85), and History (2 credits, $w_i = 2$, grade 95). Calculate the student's WGPA.

1. Calculate Weighted Grade $\sum w_i x_i$:

- Math: $4 \times 90 = 360$
- English: $3 \times 85 = 255$
- History: $2 \times 95 = 190$

$$\sum w_i x_i = 360 + 255 + 190 = 805$$

2. Total Weight $\sum w_i$:

$$\sum w_i = 4 + 3 + 2 = 9$$

3. Weighted Average (WGPA):

$$\text{WGPA} = \frac{805}{9} \approx 89.44$$

The student's Weighted Grade Point Average is approximately 89.44.



Example 12: Weighted Average Weight

Problem: A furniture store sells 3 types of chairs: Standard (weight 10 kg, 50 units sold), Premium (weight 15 kg, 30 units sold), and Basic (weight 8 kg, 20 units sold). Find the weighted average weight of a chair sold. (The number of units sold serves as the weight w_i).

1. **Calculate Total Weighted Weight** $\sum w_i x_i$:

- Standard: $50 \times 10 = 500$ kg
- Premium: $30 \times 15 = 450$ kg
- Basic: $20 \times 8 = 160$ kg

$$\sum w_i x_i = 500 + 450 + 160 = 1110 \text{ kg}$$

2. **Total Weight (Total Units Sold)** $\sum w_i$:

$$\sum w_i = 50 + 30 + 20 = 100$$

3. **Weighted Average Weight:**

$$\text{Weighted Average} = \frac{1110}{100} = 11.1$$

The weighted average weight of a chair sold is **11.1 kg**.